SEVEN EQUIVALENT FORMULATIONS OF ETALE LOCALNESS ON THE SOURCE-AND-TARGET

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 2 navigation stubs for the definition and the equivalence lemma whose numbered clauses the proof cites; each application of a clause is written out in the span.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Editorial note. The label lemma-local-source-target-characterize is defined only in descent.tex, and its canonical tag is not among the tags of the published items of this file, so it is not a published duplicate. In particular it is not the same proposition as the published item 634 (tag 04R2, 'Separate etale locality yields source-and-target locality'), which is a different labelled statement of this chapter.

Allowed standard prerequisites (editorial).

- Definition of etale localness on the source-and-target. Same chapter, Definition definition-local-source-target, invoked at proof lines 7502-7503 and 7513-7514. Supplies the definitional equivalence and clause (1).
- Equivalent formulations of etale localness and their clause numbering. Same chapter, Lemma lemma-local-source-target-implies parts (1), (2), (3) and (4), invoked at proof lines 7504-7507 and 7515-7524. Supplies the covering formulations and the transfers of the property.

Source reference. Original lemma 04R0: sources/stacks-descent/pinned-original-source.tex, source lines 7399–7399. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let $\mathcal{P}$ be a property of morphisms of schemes. We say $\mathcal{P}$ is *étale local on source-and-target* if

(1) (stable under precomposing with étale maps) if $f : X \to Y$ is étale and $g : Y \to Z$ has $\mathcal{P}$, then $g \circ f$ has $\mathcal{P}$,
(2) (stable under étale base change) if $f : X \to Y$ has $\mathcal{P}$ and $Y' \to Y$ is étale, then the base change $f' : Y' \times_Y X \to Y'$ has $\mathcal{P}$, and
(3) (locality) given a morphism $f : X \to Y$ the following are equivalent
    (a) f has $\mathcal{P}$,

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(b) for every $x \in X$ there exists a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{h} & V \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ X & \xrightarrow{f} & Y \end{array}$$

with étale vertical arrows and $u \in U$ with $a(u) = x$ such that h has $\mathcal{P}$.

Lemma 0.2. *Let $\mathcal{P}$ be a property of morphisms of schemes which is étale local on source-and-target. Let $f : X \to Y$ be a morphism of schemes. The following are equivalent:*

(a) *f has property $\mathcal{P}$,*
(b) *for every $x \in X$ there exists an étale morphism of germs $a : (U, u) \to (X, x)$, an étale morphism $b : V \to Y$, and a morphism $h : U \to V$ such that $f \circ a = b \circ h$ and h has $\mathcal{P}$,*
(c) *for any commutative diagram*

$$\begin{array}{ccc} U & \xrightarrow{h} & V \\ {\scriptstyle a}\downarrow & & \downarrow{\scriptstyle b} \\ X & \xrightarrow{f} & Y \end{array}$$

with a, b étale the morphism h has $\mathcal{P}$,
(d) *for some diagram as in (c) with $a : U \to X$ surjective h has $\mathcal{P}$,*
(e) *there exists an étale covering $\{Y_i \to Y\}_{i \in I}$ such that each base change $Y_i \times_Y X \to Y_i$ has $\mathcal{P}$,*
(f) *there exists an étale covering $\{X_i \to X\}_{i \in I}$ such that each composition $X_i \to Y$ has $\mathcal{P}$,*
(g) *there exists an étale covering $\{Y_i \to Y\}_{i \in I}$ and for each $i \in I$ an étale covering $\{X_{ij} \to Y_i \times_Y X\}_{j \in J_i}$ such that each morphism $X_{ij} \to Y_i$ has $\mathcal{P}$.*

Proof. The equivalence of (a) and (b) is part of Definition 0.1. The equivalence of (a) and (e) is Lemma 04R0 part (2). The equivalence of (a) and (f) is Lemma 04R0 part (1). As (a) is now equivalent to (e) and (f) it follows that (a) equivalent to (g).

It is clear that (c) implies (a). If (a) holds, then for any diagram as in (c) the morphism $f \circ a$ has $\mathcal{P}$ by Definition 0.1 part (1), whereupon h has $\mathcal{P}$ by Lemma 04R0 part (4). Thus (a) and (c) are equivalent. It is clear that (c) implies (d). To see that (d) implies (a) assume we have a diagram as in (c) with $a : U \to X$ surjective and h having $\mathcal{P}$. Then $b \circ h$ has $\mathcal{P}$ by Lemma 04R0 part (3). Since $\{a : U \to X\}$ is an étale covering we conclude that f has $\mathcal{P}$ by Lemma 04R0 part (1). □